Paired Matched Terminal Supervision and Direction-Gain Allocation for Budgeted Sample-Column Interventions

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***Abstract*—Tabular models can use schema information, such as column names and data types, to interpret cell values, yet a uniform schema-semantic strategy may not benefit all samples equally. Under a limited intervention budget, sample-column interventions require within-sample action selection and cross-row budget allocation, but reliable supervision for estimating incremental value is lacking. We introduce Paired matched terminal supervision, deriving intervention credit from matched terminal outcomes: PairedMagnitude preserves continuous gain, whereas PairedSign preserves beneficial direction. Sign-to-Magnitude uses PairedSign for within-row candidate selection and a PairedMagnitude estimator for cross-row allocation. In the main supervision comparison, PairedMagnitude achieves the highest mean Gain at all seven budgets, reaching 0.003256 at five percent. At five percent under dataset-equal aggregation, Sign-to-Magnitude achieves higher mean Gain and F1 change than Sign-only and, relative to Magnitude-only, lower Harm of 29.86 percent versus 42.98 percent and higher mean F1 change of 0.001606 versus 0.001000, despite lower mean Gain. These results support the value of matched intervention credit and stage-specific use of direction and magnitude information.**

Keywords—paired supervision, matched terminal supervision, budgeted intervention, sample-column ranking, Sign-to-Magnitude

I. INTRODUCTION

Tabular data are widely used in applications such as financial risk assessment, medical prediction, e-commerce, and industrial analytics. In many tabular models, cell interpretation can depend on both observed values and schema information, such as column names and data types [1]. However, the same schema-semantic strategy may benefit some samples but be ineffective or harmful for others within the same column. Row-level methods can prioritize samples but do not specify which column action should change, while column-level strategies cannot capture sample-specific differences. This motivates fine-grained sample-column interventions. Under a limited budget, such interventions require selecting one candidate within each sample and allocating budget across the retained candidates.

Existing downstream-aware data processing methods select records, features, or cells through cleaning, repair, confirmation, specialized processing, or retraining [2]–[5]. Effect-ranking, uplift, and learning-to-rank methods prioritize interventions by estimated incremental benefit under resource constraints [6]–[11]. These methods use different action spaces and feedback assumptions, leaving open how to construct traceable incremental terminal credit for the controlled sample-column intervention studied here. Directly learning an intervention's absolute terminal outcome may also mix its local effect with sample-specific difficulty and other unchanged decisions.

We introduce Paired matched terminal supervision, which constructs intervention credit from the terminal outcomes of the same sample under the default (Static) strategy and a single-column alternative (Complement). PairedMagnitude preserves continuous gain information, while PairedSign preserves beneficial direction information. We further propose Sign-to-Magnitude, which uses PairedSign for within-row candidate selection and fits a dedicated PairedMagnitude estimator on the selected winners for cross-row budget allocation.

The contributions of this paper are summarized as follows:

(1) Paired matched terminal supervision. We construct matched sample-column incremental credit with PairedMagnitude and PairedSign representations.

(2) Sign-to-Magnitude allocation. We use PairedSign for within-row selection and a PairedMagnitude estimator for cross-row allocation.

II. RELATED WORK

A. Downstream-Aware Cleaning

ActiveClean prioritizes records by model gradients [2], CPClean resolves uncertainty through truth revelation [4], CDC evaluates cleaning by downstream performance [3], and COMET recommends cost-aware feature-wise actions [5]. These works establish downstream-aware prioritization across

different action granularities. Our study instead considers budgeted sample-column schema-semantic interventions.

B. Effect Ranking and Learning to Rank

Treatment-effect ranking and uplift methods prioritize units by incremental rather than absolute outcomes. Causal forests estimate heterogeneous effects [9]; direct ranking and constrained allocation methods optimize effect-based prioritization [8], [10], while rank-weighted estimands evaluate prioritization rules [11]. PairLogit [6] and LambdaMART [7] serve as alternative learning-to-rank models for candidate ordering. For controlled schema interventions, we construct candidate-level terminal credit from matched Static–Complement executions and study how its direction and magnitude guide the two decision stages.

C. Decision and Evaluation Layers

Normalized discounted cumulative gain (NDCG) and pairwise accuracy measure within-row candidate ordering [6], [12]. Budgeted gain and Harm characterize the executed intervention set, while task-adapted F1 measures downstream classification change. We report these endpoints separately because candidate ranking, budget allocation, negative-credit frequency, and prediction change capture different properties.

III. Method

A. Decision Unit and Budget Protocol

Let r denote a raw sample and $C(r)$ its eligible candidate columns. Each pair (r, c) defines a sample-column intervention that toggles only the schema-semantic contribution of column c while leaving the observed value and all non-candidate column decisions unchanged; Static follows the default schema-semantic state vector, whereas Complement reverses the state of the candidate column. Static states are determined on inner validation: a column is ON when enabling it from the all-OFF state lowers mean validation NLL, and OFF otherwise. ON injects the column-name/type embedding into the column token, while OFF zeros that contribution but retains value, type, and missingness inputs. For example, when *occupation* is ON under Static for an Adult sample, Complement switches its schema embedding off while leaving the observed occupation value unchanged. Because the downstream Transformer is nonlinear, the same schema embedding may affect samples differently. Eligible columns exclude the target and require finite inner-validation ON/OFF evidence; missing cells remain eligible.

Given a budget of B% of eligible samples, Stage 1 selects the highest-scoring intervention within each eligible sample, yielding at most one row winner. Stage 2 ranks these winners across samples and executes the top B%, with the action count rounded up within each experimental cell; all remaining samples retain Static. Restricting each sample to one winner matches the single-column intervention used to define local credit. All eligible interventions are treated as equal-cost actions.

B. Paired Matched Terminal Supervision

For each eligible sample-column pair (r, c), we construct matched terminal credit by comparing Static with the corresponding single-column Complement path:

$$g(r,c)=\mathrm{NLL}_{\mathrm{Static}}(r)-\mathrm{NLL}_{\mathrm{Complement}}(r,c). \quad (1)$$

Here, the two terms are the terminal true-class negative log-likelihoods under the matched Static and Complement paths. The paths share the same raw sample, preprocessing state and corruption realization, downstream checkpoint, and all non-candidate decisions; only the candidate column state differs. Positive $g(r, c)$ indicates that Complement reduces terminal NLL, while negative credit indicates the opposite. Matched subtraction focuses on the candidate intervention's incremental terminal change rather than its absolute outcome. These NLLs come from the same frozen Transformer classifier implemented in PyTorch [13]; neither arm retrains it.

PairedMagnitude uses the continuous signed credit $g(r, c)$, whereas PairedSign uses $1[g(r, c) > \varepsilon]$, with $\varepsilon = 10^{-6}$, to retain beneficial direction. They are two representations of the same Paired matched terminal supervision.

C. Supervision Controls and Learning Carriers

We distinguish supervision construction from learning carriers and decision policies. The primary supervision comparison fixes CatBoost as the learning carrier for learned constructions and compares supervision constructions, while separate carrier experiments examine whether the observed behavior depends on a particular learner. Policy scorers use only outcome-free row values and missingness, candidate identity/type, Static state, regime indicators, and outer-training summaries; encoders and statistics are fit on outer-training data.

The supervision controls include Unpaired, Two-outcome, ColumnMean, and TargetPermutation. They respectively test supervision without a row-matched Static reference, separate outcome modeling, a column-level prior, and broken feature-credit correspondence; Table I summarizes their training targets and implementations.

TABLE I. Supervision Constructions (Fixed CatBoost [14] Carrier)

Supervision	Training target	Role
PairedMagnitude	$g(r,c)$	Signed continuous credit
PairedSign	$1[g(r,c)>\varepsilon]$	Beneficial direction
Unpaired	$-\mathrm{NLL}_{\mathrm{Complement}}(r, c)$	No row-matched Static reference
Two-outcome	Static NLL; Complement NLL (Separate Models)	Separate outcome modeling; predicted difference
ColumnMean	Training-fold column-mean credit	Column-level prior
TargetPermutation	Permuted $g(r,c)$	Target-permutation negative control

[a] Learned constructions use the fixed CatBoost configuration; Two-outcome fits separate Static and Complement models, while ColumnMean is analytic.

D. Sign-to-Magnitude Direction–Gain Allocation

Paired credit contains beneficial-direction and continuous-magnitude information, motivating their use for within-row selection and cross-row allocation, respectively.

Stage 1 uses PairedSign to select one winner per sample; after the winners are frozen, Stage 2 fits a dedicated magnitude

estimator to predict their PairedMagnitude credit for cross-row allocation. Stage 2 uses a random forest [15] on outcome-free winner features and frozen Stage-1 score descriptors; it does not reselect the column.

E. Cross-Fitting and Information Control

Outer folds are grouped by raw sample, so all candidate interventions and matched paths from the same sample remain in the same fold. All downstream-model fitting, checkpoint selection, and policy fitting are confined to the outer-training partition. Within each outer-training partition, inner validation determines the Static column states and downstream checkpoint setting; the downstream model is then refitted on the full outer-training partition for held-out scoring.

At inference, both stages use only outcome-free features. Outer-test outcomes and exact outer-test intervention credits are accessed only after the final intervention set is frozen. Cross-fitting refers only to policy scoring; inner-holdout labels are reused for checkpoint selection and credit generation, so upstream credit generation is not fully out-of-fold.

TABLE II. OVERALL PIPELINE FOR PAIRED SUPERVISION AND DIRECTION–GAIN ALLOCATION

Input: Raw samples; eligible candidate columns; schema-semantic action space; outcome-free policy features; downstream training procedure; intervention budget B.	
Output: Frozen budgeted intervention set.	
Offline terminal credit construction	
1.	Determine Static states and the downstream checkpoint setting by inner validation; execute matched Static and single-column Complement paths to construct training credits.
2.	Derive PairedMagnitude and PairedSign targets from the matched training credits.
Offline policy training	
3.	Cross-fit PairedSign to obtain out-of-fold (OOF) scores for eligible outer-training candidates.
4.	Select and freeze one highest-scoring OOF training winner per eligible sample, together with its Stage-1 score descriptors.
5.	Fit the Stage-2 magnitude estimator using the frozen winners' outcome-free features, frozen Stage-1 score descriptors, and continuous PairedMagnitude targets.
6.	Refit PairedSign on all outer-training candidate records for inference.
7.	Refit and freeze the downstream model on full outer-training data using the inner-validation checkpoint setting for outer-test evaluation.
Inference and evaluation	
8.	Apply the refitted PairedSign scorer to eligible test candidates and select one highest-scoring row winner per sample.
9.	Predict winner gains with Stage 2, rank winners across samples, and select the top B% of eligible samples for Complement; all remaining samples retain Static.
10.	Freeze the intervention set, execute the selected Complement actions, and then access outer-test outcomes for evaluation.

IV. EXPERIMENTAL PROTOCOL

A. Benchmark Scope

An experimental cell is one dataset–regime–outer-fold–seed combination. We evaluate seven UCI tabular classification datasets [16]: Online Shoppers, Dry Bean, Adult, Letter Recognition, Credit Default, WDBC, and Steel Plates Faults. Each dataset is evaluated under BASE and COMPOUND settings; BASE uses no injected corruption, whereas COMPOUND combines 30% uniform masking of observed cells with 10% training-label noise. Five raw-sample-grouped outer folds, five fixed seeds (171401, 171503, 171617, 271829, and 314159), and two regimes yield 350 experimental cells. Intervention budgets are 1%, 2%, 3%, 5%, 7%, 10%, and 15%.

B. Comparison Design

We evaluate two primary aspects: supervision construction and decision strategy. For supervision construction, we fix CatBoost as the learning carrier and compare PairedMagnitude and PairedSign with Unpaired, Two-outcome, ColumnMean, and TargetPermutation. For decision strategy, we compare Sign-to-Magnitude with Sign-only and Magnitude-only. Sign-only and Magnitude-only reuse PairedSign and PairedMagnitude, respectively, in both stages.

Additional carrier analysis repeats the three-policy comparison with histogram gradient boosting (HGB) [17], CatBoost [14], PairLogit [6], [14], and LambdaMART [7], [18] as Stage-1 carriers. Random uniformly samples one eligible column per sample and allocates the budget randomly. All comparisons use the same data splits, intervention space, budgets, and evaluator.

C. Evaluation Endpoints

Budgeted Gain, the primary execution metric, averages realized Paired credit over eligible samples, assigning zero to unacted samples. Within-row ranking is evaluated by NDCG@1 using $\max(g(r,c),0)$ as relevance [12] and by pairwise accuracy [6] over pairs with $|\Delta g| > \varepsilon$. Harm is the fraction of executed interventions with $g(r,c) < -\varepsilon$. Downstream impact is measured by ΔF1 between the same frozen intervention set and Static, using positive-class F1 for binary tasks and macro-F1 for multiclass tasks.

D. Aggregation and Reporting Rules

Folds are pooled within dataset–regime–seed strata; regimes and seeds are then equally averaged within dataset, followed by equal weighting across datasets. Main results are reported as mean ± sample standard deviation across dataset means. At the 5% budget, we additionally report 95% paired bootstrap intervals from 9,999 raw-row resamples within dataset–outer-fold strata, with shared weights across regimes and seeds, for the main Gain contrasts. These intervals condition on frozen predictions and actions.

V. RESULTS

A. Paired Supervision under a Fixed CatBoost [14] Carrier

PairedMagnitude achieves the highest mean Gain at every reported budget, whereas PairedSign lowers Harm@5% and leads NDCG@1 and pairwise accuracy (Table III; Fig. 1). Two-outcome has lower mean Gain than PairedMagnitude. At 5%, the PairedMagnitude - Two-outcome and PairedMagnitude - Unpaired Gain contrasts have conditional 95% bootstrap intervals of [1.735, 2.499] and [2.771, 3.756] ($\times 10^{-3}$), respectively. TargetPermutation breaks the feature-credit correspondence and yields negative mean Gain at every reported budget. Together, these results distinguish two effects of Paired supervision: continuous matched credit favors realized Gain, while sign supervision favors within-row ranking and lower

action risk. PairedMagnitude's higher mean Δ F1@5% than PairedSign also accompanies higher Harm@5% (Table III).

TABLE III. SUPERVISION COMPARISON WITH A FIXED CATBOOST [14] CARRIER OVER 350 CELLS

Supervision	Gain@1% ($\times 10^{-3}$)	Gain@5% ($\times 10^{-3}$)	Harm@5% (%)
PairedMagnitude	**1.226 ± 1.429**	**3.256 ± 4.277**	42.98 ± 10.41
PairedSign	0.106 ± 0.193	0.426 ± 0.825	**21.98 ± 10.83**
Unpaired	−0.001 ± 0.014	−0.002 ± 0.074	54.59 ± 7.39
Two-outcome	0.582 ± 0.764	1.141 ± 1.573	48.10 ± 4.59
ColumnMean	−0.007 ± 0.104	0.018 ± 0.472	48.91 ± 6.02
TargetPermutation	−0.274 ± 0.525	−0.733 ± 1.229	53.99 ± 4.59

Supervision	NDCG@1	Pairwise	ΔF1@5% ($\times 10^{-3}$)
PairedMagnitude	0.243 ± 0.061	0.564 ± 0.034	**+1.000 ± 3.598**
PairedSign	**0.279 ± 0.088**	**0.619 ± 0.047**	+0.349 ± 0.868
Unpaired	0.202 ± 0.055	0.543 ± 0.040	+0.005 ± 0.024
Two-outcome	0.202 ± 0.055	0.543 ± 0.040	+0.677 ± 0.645
ColumnMean	0.189 ± 0.048	0.538 ± 0.023	+0.246 ± 0.402
TargetPermutation	0.195 ± 0.051	0.536 ± 0.023	−1.815 ± 3.615

[a] Entries are mean ± sample SD across seven dataset means; folds are pooled before averaging regimes and seeds. Units are given in the column headings. Bold marks the most favorable mean; lower is better for Harm.

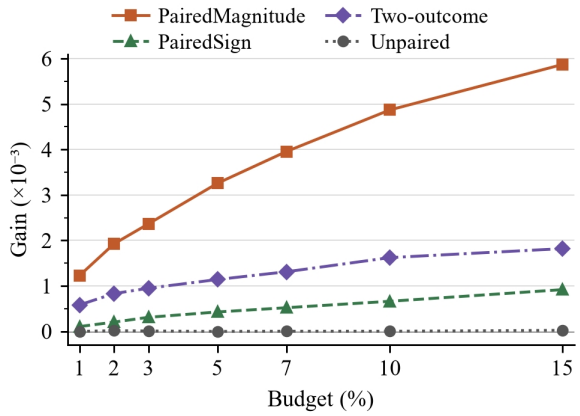


Fig. 1. Seven-budget Gain for PairedMagnitude, PairedSign, Two-outcome, and Unpaired using CatBoost [14]. Lines show seven-dataset means over the complete 350-cell scope.

B. *Sign-to-Magnitude Policy Comparison*

Under the dataset-equal aggregation, Sign-to-Magnitude at 5% trades lower mean Gain for lower Harm and higher mean ΔF1 than Magnitude-only (Table IV). The Sign-to-Magnitude − Sign-only and Sign-to-Magnitude − Magnitude-only Gain@5% contrasts have conditional 95% bootstrap intervals of [1.227, 1.796] and [−1.736, −0.912] ($\times 10^{-3}$), respectively. Magnitude-only has higher mean Gain at all seven budgets; Sign-to-Magnitude has higher mean ΔF1 at the reported budgets up to 7%, while Magnitude-only has higher mean ΔF1 at 10% and 15% (Fig. 2).

C. *Cross-Carrier Results for Sign-to-Magnitude*

Across the four carriers, the mean Gain@5% contrast between Sign-to-Magnitude and Magnitude-only is negative for HGB and CatBoost, but positive for PairLogit and LambdaMART. The mean ΔF1@5% contrast favors Sign-to-Magnitude for all four carriers (Table V); this aggregate pattern does not imply improvement on every dataset.

TABLE IV. THREE-POLICY COMPARISON OVER 350 CELLS

Policy	Gain@1% ($\times 10^{-3}$)	Gain@5% ($\times 10^{-3}$)	Harm@5% (%)
Sign-only	0.106 ± 0.193	0.426 ± 0.825	**21.98 ± 10.83**
Magnitude-only	**1.226 ± 1.429**	**3.256 ± 4.277**	42.98 ± 10.41
Sign-to-Magnitude	0.807 ± 0.924	1.935 ± 2.602	29.86 ± 7.25
Random	−0.147 ± 0.294	−0.711 ± 1.453	53.52 ± 5.28

Policy	NDCG@1	Pairwise	ΔF1@5% ($\times 10^{-3}$)
Sign-only	**0.279 ± 0.088**	**0.619 ± 0.047**	+0.349 ± 0.868
Magnitude-only	0.243 ± 0.061	0.564 ± 0.034	+1.000 ± 3.598
Sign-to-Magnitude	**0.279 ± 0.088***	**0.619 ± 0.047***	**+1.606 ± 2.304**

[a] Aggregation and units follow Table III. Bold marks the most favorable mean among learned policies; lower is better for Harm. Random is the chance reference averaged over 50 registered draws.

[b] * Sign-to-Magnitude and Sign-only share identical Stage-1 winners; their NDCG and pairwise accuracy are identical by construction. Gain, Harm, and ΔF1 report their final policies under the respective Stage-2 allocation scores.

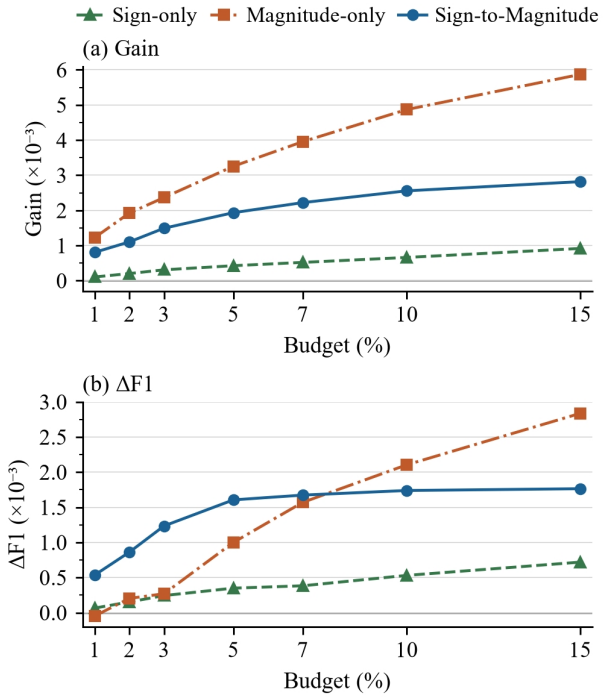


Fig. 2. Seven-budget comparison of Sign-only, Magnitude-only, and Sign-to-Magnitude over 350 cells: (a) Gain; (b) ΔF1. Lines show dataset-equal means. Both panels share the same budget points, policy markers, and aggregation.

VI. DISCUSSION

This work studies matched intervention credit for budgeted sample-column interventions. The framework provides a controlled formulation of decisions over a predefined sample-column intervention space.

TABLE V. SIGN-TO-MAGNITUDE COMPARISON ACROSS FOUR STAGE-1 CARRIERS OVER 350 CELLS

Carrier	Gain@5% ($\times 10^{-3}$)	Harm@5% (%)	ΔF1@5% ($\times 10^{-3}$)
HGB [17]	1.550 ± 1.827	29.75 ± 6.63	+1.141 ± 1.569
CatBoost [14]	**1.935 ± 2.602**	29.86 ± 7.25	+1.606 ± 2.304
PairLogit [6]	1.831 ± 2.293	30.66 ± 6.89	**+1.676 ± 2.169**
LambdaMART [7]	1.868 ± 1.910	**29.01 ± 7.35**	+1.476 ± 1.788

Carrier	ΔGain@5% ($\times 10^{-3}$): Sign-to-Magnitude− Magnitude-only	ΔF1@5% ($\times 10^{-3}$): Sign-to-Magnitude− Magnitude-only
HGB [17]	−1.827 ± 2.918	+0.996 ± 5.181
CatBoost [14]	−1.321 ± 1.735	+0.606 ± 2.863
PairLogit [6]	+1.134 ± 1.215	+1.184 ± 1.152
LambdaMART [7]	**+1.364 ± 1.409**	**+1.279 ± 1.478**

[a.] Aggregation and units follow Table III; contrasts are paired within dataset before SD calculation. Contrasts compare complete policies; Stage-1 winners may differ. Boldface marks the most favorable mean (lower for Harm).

NLL Gain and ΔF1 capture different intervention effects. NLL Gain measures changes in true-class probability, while ΔF1 depends on whether those changes alter class predictions and the resulting confusion structure. Thus, a large NLL improvement may leave F1 unchanged, whereas a smaller probability shift can affect F1 if it changes the predicted class. For example, at 5%, Magnitude-only has higher mean Gain but lower mean ΔF1 than Sign-to-Magnitude (Table IV). Harm provides a complementary view by measuring how often executed interventions worsen terminal NLL. Non-trivial Harm also motivates abstention or risk-aware allocation in higher-cost settings. The 5% mean Gains of both policies are below 1% of the dataset-equal Static NLL mean (0.423 ± 0.298 across datasets), indicating small average NLL reductions. Their practical significance depends on whether these probability-level changes improve downstream utility enough to justify intervention cost and risk. The reported SDs and conditional Gain intervals do not establish F1 significance or retraining stability.

VII. LIMITATIONS

The current study provides an initial controlled formalization of budgeted sample-column intervention, based on a predefined intervention space with single-column, equal-cost actions. Real systems may involve interacting interventions, heterogeneous costs, and additional operational constraints; extending the framework to these settings remains important.

Paired supervision requires offline matched executions to obtain terminal credit, which may limit applicability when such feedback is costly or unavailable, although inference uses only outcome-free features. Experiments are limited to benchmark tabular datasets and controlled corruption settings, so the results do not establish deployment benefit; further validation in larger and more diverse real-world settings is needed.

VIII. CONCLUSION

Paired matched terminal supervision constructs sample-column incremental credit from matched Static–Complement true-class NLL outcomes. PairedMagnitude captures continuous gain, while PairedSign captures beneficial direction. In the fixed-carrier supervision comparison, PairedMagnitude achieves the highest mean Gain at all seven budgets, whereas PairedSign yields lower Harm at 5% and stronger within-row ranking. Under dataset-equal aggregation at 5%, Sign-to-Magnitude achieves lower Harm and higher mean ΔF1 than Magnitude-only, despite lower mean Gain. These results support stage-specific use of direction and magnitude within the controlled budgeted-intervention formulation.

REFERENCES

[1] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, "TabLLM: Few-shot classification of tabular data with large language models," in Proc. 26th Int. Conf. Artificial Intelligence and Statistics (AISTATS), PMLR, vol. 206, pp. 5549–5581, 2023.

[2] S. Krishnan, J. Wang, E. Wu, M. J. Franklin, and K. Goldberg, "ActiveClean: Interactive data cleaning for statistical modeling," Proc. VLDB Endow., vol. 9, no. 12, pp. 948–959, 2016, doi: 10.14778/2994509.2994514.

[3] S. Jäger and F. Biessmann, "From data imputation to data cleaning—Automated cleaning of tabular data improves downstream predictive performance," in Proc. 27th Int. Conf. Artificial Intelligence and Statistics (AISTATS), PMLR, vol. 238, pp. 3394–3402, 2024.

[4] B. Karlaš et al., "Nearest neighbor classifiers over incomplete information: From certain answers to certain predictions," Proc. VLDB Endow., vol. 14, no. 3, pp. 255–267, 2021, doi: 10.14778/3430915.3430917.

[5] S. Mohammed, F. Naumann, and H. Harmouch, "Step-by-step data cleaning recommendations to improve ML prediction accuracy," in Proc. 28th Int. Conf. Extending Database Technology (EDBT), pp. 542–554, 2025, doi: 10.48786/edbt.2025.43.

[6] C. Burges et al., "Learning to rank using gradient descent," in Proc. 22nd Int. Conf. Machine Learning (ICML), pp. 89–96, 2005, doi: 10.1145/1102351.1102363.

[7] Q. Wu, C. J. C. Burges, K. M. Svore, and J. Gao, "Adapting boosting for information retrieval measures," Inf. Retr., vol. 13, no. 3, pp. 254–270, 2010, doi: 10.1007/s10791-009-9112-1.

[8] F. Devriendt, J. Van Belle, T. Guns, and W. Verbeke, "Learning to rank for uplift modeling," IEEE Trans. Knowl. Data Eng., vol. 34, no. 10, pp. 4888–4904, 2022, doi: 10.1109/TKDE.2020.3048510.

[9] S. Wager and S. Athey, "Estimation and inference of heterogeneous treatment effects using random forests," J. Amer. Statist. Assoc., vol. 113, no. 523, pp. 1228–1242, 2018, doi: 10.1080/01621459.2017.1319839.

[10] F. Kamran, M. Makar, and J. Wiens, "Learning to rank for optimal treatment allocation under resource constraints," in Proc. 27th Int. Conf. Artificial Intelligence and Statistics (AISTATS), PMLR, vol. 238, pp. 3727–3735, 2024.

[11] S. Yadlowsky, S. Fleming, N. Shah, E. Brunskill, and S. Wager, "Evaluating treatment prioritization rules via rank-weighted average treatment effects," J. Amer. Statist. Assoc., vol. 120, no. 549, pp. 38–51, 2025, doi: 10.1080/01621459.2024.2393466.

[12] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of IR techniques," ACM Trans. Inf. Syst., vol. 20, no. 4, pp. 422–446, 2002, doi: 10.1145/582415.582418.

[13] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in Adv. Neural Inf. Process. Syst., vol. 32, pp. 8024–8035, 2019.

[14] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in Adv. Neural Inf. Process. Syst., vol. 31, pp. 6638–6648, 2018.

[15] L. Breiman, "Random forests," Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[16] M. Kelly, R. Longjohn, and K. Nottingham, "The UCI Machine Learning Repository," 2023. [Online]. Available: https://archive.ics.uci.edu

[17] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[18] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in Adv. Neural Inf. Process. Syst., vol. 30, pp. 3146–3154, 2017.